香港中文大學(深圳)
The Chinese University of Hong Kong, Shenzhen

ECO6102 / 2026–27 Term 1 (Fall)

Lecture 3b: Linear Regression

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Lecture 3 · Monday 28 September 2026

Announcements

Office hours move; slides and Problem Set 1

- **Office hours, new time:** Monday 12:00–13:00, Xili Campus 6A204 — right after class.
- **Slides:** the no-animation version of each lecture is now on Blackboard.
- **Problem Set 1:** due Sunday 11 October 2026.

Roadmap (3b)

Uncertainty, omitted variables and measurement error

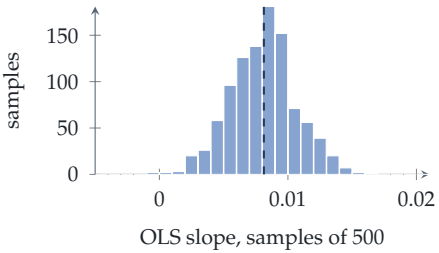
OLS Uncertainty and Inference

OLS Solutions to the Omitted Variables Problem

OLS under Measurement Error

Where we stopped

Tests need the centre, the spread and the shape of the sampling distribution



- **Centre**: $\hat{\boldsymbol{\beta}} \xrightarrow{p} \boldsymbol{\beta}$ under OLS.1–OLS.2 (unbiased if $E(u \mid \mathbf{x}) = 0$). Done in 3a.
- **Spread**: how much $\hat{\boldsymbol{\beta}}$ varies $\Rightarrow$ its variance and standard error. Today.
- **Shape**: normal, by either of two routes:
  - normal homoskedastic errors $\Rightarrow$ exact normality given $\mathbf{X}$;
  - no normality, large $N \Rightarrow$ approximately normal (Theorem 4.2).

Roadmap

Wooldridge 4.2.2–4.2.3 — current location in blue

OLS Uncertainty and Inference
- OLS.3 and Standard Errors
- Testing Hypotheses

OLS Solutions to the Omitted Variables Problem
- OLS Ignoring the Omitted Variables
- Proxy Variable–OLS Solution
- Interactions in Unobservables: Random Coefficients

OLS under Measurement Error
- Measurement Error in the Dependent Variable
- Measurement Error in an Explanatory Variable

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From sampling spread to a standard error

Homoskedasticity gives a familiar uncertainty formula

THE FAMILIAR ASSUMPTIONS

$$E(u \mid \mathbf{x}) = 0, \qquad \boxed{\mathrm{Var}(u \mid \mathbf{x}) = \sigma^2}.$$

- **Homoskedasticity**: the error has the same spread at every value of the regressors.
- **Standard error**: our estimate of how much $\hat{\beta}_j$ varies across samples.
- The assumptions above imply OLS.3, which simplifies this calculation.

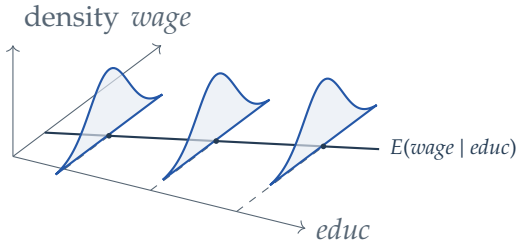
▸ How the assumptions imply OLS.3 ▸ From variance to standard error

Does the error have the same spread at every schooling level?

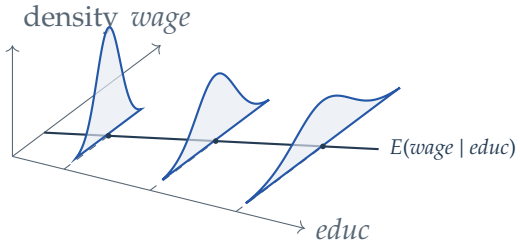
▸ Standard errors in the attendance example

Homoskedasticity in a picture

The mean line stays fixed; the spread changes



homoskedastic: constant spread



heteroskedastic: spread grows with *educ*

Wages at $educ = 8$ and at $educ = 16$: the same spread, or a wider one at 16? Which picture, and why?

Why wages spread out with schooling

A wider range of jobs can produce a wider range of wages

$$wage = \beta_0 + \beta_1 educ + u, \qquad E(u \mid educ) = 0.$$

- More schooling may open a wider range of jobs and wages.
- With little schooling, wages may cluster near the minimum wage.
- A changing spread makes constant conditional variance doubtful.

How can one sample tell us about an unknown population coefficient?

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From a sample to the population

Inference concerns the unknown population coefficient

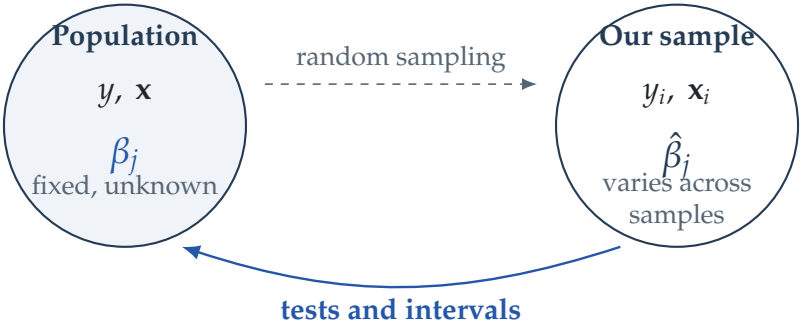
- **Statistical inference**: use sample data to draw conclusions about population parameters.

Population	**Our sample**
β_j: fixed but unknown	$\hat{\beta}_j$: an estimate
	Changes from sample to sample

- Hypothesis testing is based on *population* parameters, using estimate from a particular *sample*.

Inference in one picture

The sample gives $\hat{\beta}_j$; inference reasons back to β_j



What would we expect to see if $\beta_j = 0$?

The null hypothesis

Does this regressor have a nonzero partial effect?

$$y = \beta_0 + \beta_1 x_1 + \cdots + \beta_j x_j + \cdots + \beta_K x_K + u \qquad (4.1)$$

$$H_0 : \beta_j = 0, \qquad H_1 : \beta_j \neq 0 \quad \text{(two-sided alternative)}.$$

- H_0: once the other regressors are accounted for, x_j has no effect on $E(y \mid \mathbf{x})$.
- We observe $\hat{\beta}_j$, which is rarely 0. **How far is $\hat{\beta}_j$ from zero?**
  - If $\hat{\beta}_j$ is very far, then it's evidence against null H_0
  - But $\hat{\beta}_j$ varies also because of sampling errors! $\Rightarrow$ have to take that into account.

▸ One-sided alternatives

The t statistic

Measure the estimate in standard-error units

$$H_0 : \beta_j = 0 : \qquad \hat{t}_j = \frac{\hat{\beta}_j}{\text{se}(\hat{\beta}_j)}.$$

- $\hat{t}_j$ is the distance from zero in estimated standard deviations.
- A large $|\hat{t}_j|$ is evidence against H_0: reject when $|\hat{t}_j| > c$.
- Under H_0: approximately $N(0,1)$ under OLS.1–OLS.3 in large samples; exact t_{N-K} under the normal homoskedastic model.

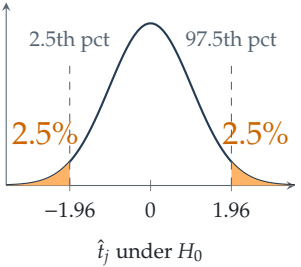
▸ The t distribution ▸ Testing a nonzero value

How far from zero should the rejection cutoff be?

Type I error

Rejecting a true null hypothesis

- **Type I error**: reject H_0 when it is true; $\Pr(\text{reject } H_0 \mid H_0 \text{ true}) \approx \alpha$.
- $\alpha = .05$: if H_0 is true, about 5 samples in 100 still reject.
- 2.5% per tail: $-c$ and c are the 2.5th and 97.5th percentiles.
- Smaller α or few degrees of freedom: larger c (1.645, 1.96, 2.576 at 10, 5, 1%).



▸ Table of critical values

Type II error

Failing to reject a false null hypothesis

- **Type II error**: fail to reject H_0 when it is false.
- A coefficient can be nonzero even when this sample does not detect it.
- **Power**: the probability of rejecting H_0 under a specified alternative.

$$\text{Power} = 1 - \Pr(\text{Type II error}).$$

▸ Sample size and power

Two possible errors

Our decision and the population truth are different

	Population truth	
Our decision	H_0 true	H_0 false
Reject H_0	Type I error	Correct
Fail to reject H_0	Correct	Type II error

Example 4.1: the wage equation

Do young children predict working married women's wages?

$$\log(wage) = \beta_0 + \beta_1 exper + \beta_2 exper^2 + \beta_3 educ + \beta_4 age + \beta_5 kidslt6 + \beta_6 kidsge6 + u. \quad (4.16)$$

- *wage*: hourly wage; *exper*: years of work experience.
- *educ*: years of schooling; *age*: age in years.
- *kidslt6* and *kidsge6*: children under six and six or older.

Example 4.1: the young-children coefficient

One coefficient, one question

$$\log(wage) = \beta_0 + \beta_1 exper + \beta_2 exper^2 + \beta_3 educ + \beta_4 age + \beta_5 kidslt6 + \beta_6 kidsge6 + u. \qquad (4.16)$$

- β_5: holding the other regressors fixed, one more child under six is associated with a wage difference of about $100 \cdot \beta_5$ percent.
- Does it differ from zero?

$$H_0 : \beta_5 = 0, \qquad H_1 : \beta_5 \neq 0 \quad \text{(two-sided)}.$$

Example 4.1: the estimates

The coefficients and their standard errors

	Coefficient	S.E.
Intercept	−.421	.317
exper	.040	.013
$exper^2$	−.00078	.00040
educ	.108	.014
age	−.0015	.0053
kidslt6	**−.061**	**.089**
kidsge6	−.015	.028
$N = 428$	$R^2 = .158$	

Example 4.1: one coefficient test

The estimate is less than one standard error from zero

TARGET COEFFICIENT

$$\log(wage) = \beta_0 + \cdots + \beta_5\, kidslt6 + \beta_6 kidsge6 + u, \qquad H_0 : \beta_5 = 0$$

$$\hat{t}_5 = \frac{-.061}{.089} \approx -.69$$

- $|\hat{t}_5| = .69 < 1.96$: fail to reject H_0 at 5%.
- This sample does not establish that the partial effect differs from zero.

Can we report the evidence without choosing just one significance cutoff?

Why report a p-value?

The smallest significance level at which we would reject

- The choice of α is arbitrary: researchers use 10%, 5% or 1%; there is no "correct" level.
- "Not rejected at 5%" hides information: the same result may be rejected at 10%.
- ***p*-value**: given the observed $\hat{t}_j$, the smallest significance level at which H_0 would be rejected.
- Equivalently: the probability of a t statistic at least as extreme as ours if H_0 is true.
- Small p: evidence against H_0. Large p: little evidence. With $p = .50$, half of all samples would give a t this extreme when H_0 is true.

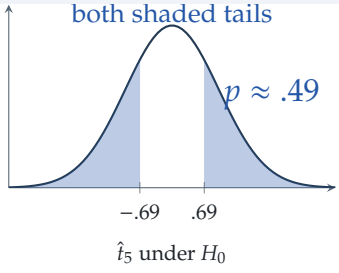
The p-value for young children

How unusual is this result if the null is true?

THE T TEST

$H_0 : \beta_5 = 0, \qquad \hat{t}_5 = -.061/.089 \approx -.69$

$$p = \Pr(|Z| \geq |\hat{t}_5|), \qquad Z \sim N(0,1).$$



- Here $p \approx .49$.
- If $\beta_5 = 0$, a $|\hat{t}_5|$ of .69 or more occurs in about 49% of samples: weak evidence against H_0.
- Reject when $p < \alpha$.

▸ Two ways to run the test

Confidence intervals

An interval of plausible values built from three numbers

$$\hat{\beta}_j \pm c \cdot \text{se}(\hat{\beta}_j), \qquad c \approx 1.96 \quad \text{for a large-sample 95\% interval.}$$

- Three ingredients: $\hat{\beta}_j$, $\text{se}(\hat{\beta}_j)$ and c, chosen for the confidence level and reference distribution. Roughly: $\hat{\beta}_j \pm 2\,\text{se}(\hat{\beta}_j)$.
- **Meaning**: if samples were drawn over and over, the intervals would contain the fixed β_j in approximately 95% of them. For our one sample we cannot know.
- **A test**: reject $H_0 : \beta_j = a_j$ at 5% exactly when a_j lies outside the interval.

▸ What intervals and tests assume

Example 4.1: an interval for young children

Insignificant does not mean economically small

YOUNG CHILDREN COEFFICIENT

$\hat{\beta}_5 = -.061, \qquad \text{se}(\hat{\beta}_5) = .089$

$$-.061 \pm 1.96(.089) = [-.235,\ .113].$$

- Zero lies inside: fail to reject $H_0 : \beta_5 = 0$, as with the t test.
- So do sizeable effects in both directions. The sample cannot rule out an economically important effect.

‣ Tests for other wage coefficients ‣ Statistical and economic significance

What if our question concerns several coefficients together?

Testing several coefficients at once

A multiple, or joint, hypotheses test

EXAMPLE 4.1

$$\log(wage) = \beta_0 + \beta_1 exper + \beta_2 exper^2 + \beta_3 educ + \beta_4 age + \beta_5 kidslt6 + \beta_6 kidsge6 + u$$

- **Question**: once experience and schooling are controlled for, do age and children matter *at all*?

$$H_0 : \beta_4 = \beta_5 = \beta_6 = 0, \qquad H_1 : \text{at least one is nonzero.}$$

- Several restrictions tested together: a **multiple (joint) hypotheses test**.
- Three separate t tests do not answer it: correlated regressors can each look insignificant yet matter together.

▸ Why separate tests can disagree with a joint test

Restricted and unrestricted models

Drop the tested group and see how much worse the fit gets

$$\begin{aligned} \text{Unrestricted:} \quad \log(wage) &= \beta_0 + \beta_1 exper + \beta_2 exper^2 + \beta_3 educ \\ &\quad + \beta_4 age + \beta_5 kidslt6 + \beta_6 kidsge6 + u \\ \text{Restricted } (H_0): \quad \log(wage) &= \beta_0 + \beta_1 exper + \beta_2 exper^2 + \beta_3 educ + u \end{aligned}$$

- Dropping regressors can only raise the sum of squared residuals: $SSR_r \geq SSR_{ur}$.
- In MROZ: $SSR_{ur} = 187.988$ and $SSR_r = 188.305$, a rise of only about .32.

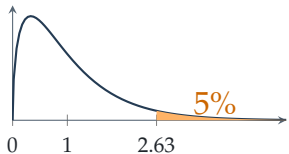
How large must the rise in residual variation be to reject the joint null?

The F statistic

The relative rise in the SSR, per restriction

$$F = \frac{(SSR_r - SSR_{ur})/Q}{SSR_{ur}/(N-K)}, \qquad Q = \text{number of restrictions.}$$

- **In words**: a large rise in the SSR when the group is dropped says the group matters.
- Under H_0, use $F_{Q,N-K}$ as an approximate reference in large samples; exact with normal homoskedastic errors.
- Large F rejects: upper tail only.



$F_{3,421}$ under H_0

‣ One restriction: t and F agree ‣ The large-sample Wald version

Example 4.1: the joint decision

The sample does not establish an additional age or children effect

JOINT NULL

$H_0 : \beta_4 = \beta_5 = \beta_6 = 0, \qquad Q = 3, \qquad N - K = 421 \ (K = 7)$

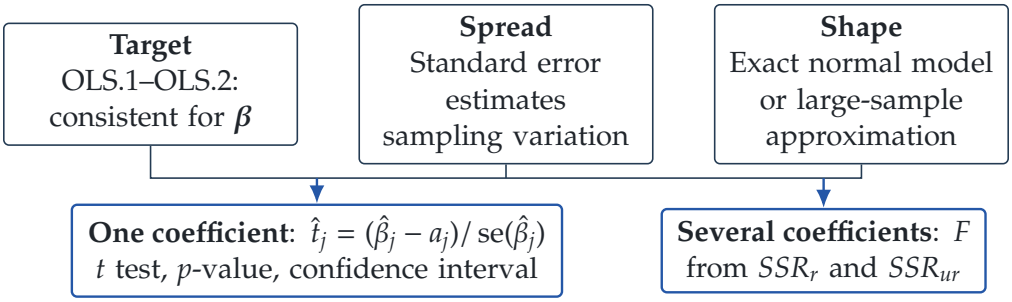
$$F = \frac{(188.305 - 187.988)/3}{187.988/421} \approx .24, \qquad p \approx .87$$

- $F = .24 < 2.63$, the 5% critical value: fail to reject.
- Dropping a group is an **exclusion restriction**; testing exclusion restrictions is the main use of F.
- Many correlated measures of one idea (firm performance in CEO pay): test them as a group.

▸ Which test for which hypothesis

From the sampling distribution to the tests

The sampling distribution supplies the reference for inference



Can a precisely estimated coefficient still be the wrong effect?

Roadmap

Wooldridge 4.3 — current location in blue

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The omitted-variable model

The target holds the unobserved factor fixed

$$E(y \mid \mathbf{x}, q) = \beta_0 + \beta_1 x_1 + \cdots + \beta_K x_K + \gamma q. \tag{4.18}$$

- q is an unobserved determinant of y.
- We want β_j: the partial effect of x_j, holding the other regressors *and* q fixed.

The schooling effect holds ability fixed

The wage equation gives each term an economic meaning

$$E(\log(wage) \mid exper, educ, abil) = \beta_0 + \beta_1 exper + \beta_2 exper^2 + \beta_3\, educ + \gamma\, abil.$$

- Schooling effect: compare workers with the same experience and ability.
- Ability contribution: ability can affect wages even though we cannot observe it.

If ability is missing from our data, where does its effect go?

Omitting a variable can create endogeneity

Its effect remains in the error of the observed equation

$$y = \beta_0 + \beta_1 x_1 + \cdots + \beta_K x_K + \gamma q + v,$$
$$E(v \mid \mathbf{x}, q) = 0. \tag{4.19–4.20}$$

v is the structural error: what remains after including q.

$$y = \beta_0 + \beta_1 x_1 + \cdots + \beta_K x_K + u, \qquad u = \gamma q + v. \tag{4.21–4.22}$$

$$\operatorname{Cov}(x_j, u) = \gamma \operatorname{Cov}(x_j, q).$$

If q affects y and is correlated with x_j, then x_j is endogenous.

▸ Recall: the structural target from Lecture 3a

Where the omitted contribution goes

Keep the original model in view while substituting for the omitted factor

$$\text{Model:} \quad y = \beta_0 + \sum_{j=1}^{K} \beta_j x_j + \gamma q + v.$$

$$\text{Projection:} \quad q = \delta_0 + \sum_{j=1}^{K} \delta_j x_j + r. \qquad (4.23)$$

$$\text{Substitute:} \quad y = (\beta_0 + \gamma\delta_0) + \sum_{j=1}^{K} (\beta_j + \gamma\delta_j) x_j + (v + \gamma r).$$

Blue: target effect. Red: omitted contribution.

Which coefficient will OLS estimate when we leave q out?

What is omitted-variable bias?

OLS converges to the composite slope instead of the target

REWRITTEN REGRESSION

$$y = (\beta_0 + \gamma\delta_0) + \sum_{j=1}^{K}(\beta_j + \gamma\delta_j)x_j + (v + \gamma r)$$

- The new error has mean zero and is uncorrelated with every regressor: OLS.1 holds for the composite coefficients.
- With OLS.2 and random sampling, Theorem 4.1 gives

$$\operatorname{plim} \hat{\beta}_j = \beta_j + \gamma\delta_j.$$

- **Large-sample omitted-variable bias**:

$$\operatorname{plim} \hat{\beta}_j - \beta_j = \gamma\delta_j.$$

▸ Why the textbook uses consistency here

Two links determine the direction

An omitted factor must affect the outcome and be related to the regressor

SPECIAL CASE IN THE TEXTBOOK

$q = \delta_0 + \delta_K x_K + r, \quad \text{Cov}(x_j, r) = 0 \ (j = 1, \ldots, K)$

$$\text{plim}\, \hat{\beta}_K = \beta_K + \gamma\delta_K = \beta_K + \gamma \frac{\text{Cov}(x_K, q)}{\text{Var}(x_K)}. \tag{4.24}$$

1. γ: does the omitted factor q raise or lower y, holding $\mathbf{x}$ fixed?
2. δ_K: do people with higher x_K also tend to have higher q?

Two positive links give an upward distortion. In the general model, δ_j is a partial association.

Omitted ability: the setup

Assume schooling captures the ability component related to the regressors

$$\log(wage) = \beta_0 + \beta_1 exper + \beta_2 exper^2 + \beta_3 educ + \gamma abil + v.$$
$$E(v \mid exper, educ, abil) = 0.$$

$$abil = \delta_0 + \delta_3 educ + r.$$

Assume r has mean zero and is uncorrelated with $exper$, $exper^2$ and $educ$.

$$\log(wage) = (\beta_0 + \gamma\delta_0) + \beta_1 exper + \beta_2 exper^2 + (\beta_3 + \gamma\delta_3) educ + (v + \gamma r).$$

Why the experience coefficients survive

The assumed ability projection contributes nothing to these slopes

GIVEN

$abil = \delta_0 + \delta_3 educ + r$, $\quad r$ is uncorrelated with all regressors

$$\log(wage) = (\beta_0 + \gamma\delta_0) + \beta_1 exper + \beta_2 exper^2 + (\beta_3 + \gamma\delta_3) educ + (v + \gamma r).$$

$$\operatorname{plim} \hat{\beta}_1 = \beta_1 + \gamma \cdot 0 = \beta_1, \qquad \operatorname{plim} \hat{\beta}_2 = \beta_2 + \gamma \cdot 0 = \beta_2.$$

Under this assumption, the experience coefficients are consistently estimated: their population coefficients did not change.

Why the schooling coefficient is too high

Positive ability effects and positive sorting add an upward contribution

$$abil = \delta_0 + \delta_3 educ + r, \qquad \gamma > 0, \quad \delta_3 > 0$$

$$\log(wage) = \cdots + (\beta_3 + \gamma\delta_3) educ + (v + \gamma r).$$

$$\operatorname{plim} \hat{\beta}_3 = \beta_3 + \gamma\delta_3 > \beta_3.$$

1. $\gamma > 0$: higher ability raises wages, holding schooling fixed.
2. $\delta_3 > 0$: higher-ability workers tend to have more schooling.

If ability is unobserved, can an observed proxy help us hold it fixed?

‣ Revisiting quiz 1: AI and pay

Roadmap

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Using IQ as a proxy for unobserved ability

A proxy provides information about the factor we cannot observe

THE SAME STRUCTURAL MODEL

$$\log(wage) = \beta_0 + \beta_3 educ + \text{other controls} + \gamma abil + v.$$

- **x**: schooling and other observed controls; q: unobserved ability.
- z: observed IQ, our proposed proxy; v: the structural error.
- A proxy need not measure ability perfectly. It must remove the ability component still related to schooling and the controls.

▸ Map of the equations and symbols ▸ Imperfect and unsuitable proxies

If true ability were already known, should IQ still help explain wages?

Redundancy

The proxy adds nothing once the underlying factor is known

$$E(y \mid \mathbf{x}, q, z) = E(y \mid \mathbf{x}, q). \tag{4.25}$$

- **Wage example**: given ability and the observed controls, IQ adds nothing to mean log wages.

▸ Why redundancy gives error orthogonality

How much of ability must IQ capture for the schooling coefficient to be consistent?

Sufficiency

The proxy must capture the part of ability related to the regressors

$$L(q \mid 1, \mathbf{x}, z) = L(q \mid 1, z). \tag{4.26}$$

$$q = \underbrace{\theta_0 + \theta_1 z}_{\text{captured by the proxy}} + \underbrace{r}_{\text{left over}}. \tag{4.27}$$

- **By construction**: $E(r) = 0$ and $\text{Cov}(z, r) = 0$.
- **Additional assumption**: $\text{Cov}(x_j, r) = 0$ for every j.

IQ need not measure ability perfectly. The remaining ability must no longer be linearly related to schooling or the other regressors.

IQ as a proxy for ability: the setup

Example 4.3: the target is the schooling effect holding ability fixed

$$\log(wage) = \beta_0 + \beta_1 exper + \beta_2 tenure + \beta_3 married + \beta_4 south + \beta_5 urban + \beta_6 black + \beta_7 educ + \gamma abil + v. \tag{4.29}$$

- Indicators: *married*, married; *south*, southern region; *urban*, metropolitan area; *black*, Black.
- We observe *IQ*, not *abil*. We assume IQ satisfies the proxy conditions, next.

The assumptions that make IQ useful

The two proxy conditions apply to different parts of the wage error

$$\log(wage) = \beta_0 + \cdots + \beta_7 educ + \gamma abil + v,$$
$$abil = \theta_0 + \theta_1 IQ + r.$$

1. **Redundancy**: once true ability and the controls are fixed, IQ adds nothing to mean log wages. Thus $E(IQ \cdot v) = 0$.
2. **Sufficiency**: remaining ability r is uncorrelated with schooling and the other controls.

The new error $\gamma r + v$ is uncorrelated with the included regressors, including IQ. With full rank, Theorem 4.1 gives consistency.

‣ Check the error correlations ‣ The general proxy substitution

Before seeing the estimates

We add observed IQ to the feasible wage regression

$$\log(wage) = a_0 + b_1 exper + b_2 tenure + b_3 married + b_4 south + b_5 urban + b_6 black + b_7 educ + \lambda IQ + \varepsilon.$$

First omit IQ; then include it, using the same workers and other regressors. Without IQ, schooling also picks up a positive ability contribution.

If IQ removes that contribution, should the schooling coefficient rise or fall?

Adding IQ lowers the schooling coefficient

Example 4.3: compare the same workers before and after adding the proxy

	Without IQ		**With IQ**	
	Coefficient	S.E.	Coefficient	S.E.
Intercept	5.395	.113	5.176	.128
exper	.014	.003	.014	.003
tenure	.012	.002	.011	.002
married	.199	.039	.200	.039
south	-.091	.026	-.080	.026
urban	.184	.027	.182	.027
black	-.188	.038	-.143	.039
educ	**.065**	.006	**.054**	.007
IQ			.0036	.0010

▸ Other readings of the IQ table

Why the schooling coefficient falls

A valid proxy removes the omitted ability contribution

$$\log(wage) = \beta_0 + \cdots + \beta_7 educ + \gamma abil + v$$

$$\text{Without IQ:} \quad \operatorname{plim} \hat{\beta}_{7,\text{omit}} = \beta_7 + \gamma\delta_7,$$

$$\text{Valid IQ proxy:} \quad \operatorname{plim} \hat{\beta}_{7,\text{proxy}} = \beta_7.$$

1. $\gamma > 0$: ability raises wages, holding the controls fixed.
2. $\delta_7 > 0$: schooling and ability have a positive partial association.

▸ A lagged outcome as a proxy

Could ability also change the return to schooling, rather than only the wage level?

Roadmap

Wooldridge 4.3.3 — current location in blue

OLS Uncertainty and Inference
- OLS.3 and Standard Errors
- Testing Hypotheses

OLS Solutions to the Omitted Variables Problem
- OLS Ignoring the Omitted Variables
- Proxy Variable–OLS Solution
- **Interactions in Unobservables: Random Coefficients**

OLS under Measurement Error
- Measurement Error in the Dependent Variable
- Measurement Error in an Explanatory Variable

Interactions in unobservables

The effect of an observed regressor can depend on an unobserved characteristic

In some cases we might be concerned about interactions between unobservables and observable regressors.

$$y = \beta_0 + \beta_1 x_1 + \cdots + \beta_K x_K + \gamma_1 q + \gamma_2 x_K q + v, \qquad E(v \mid \mathbf{x}, q) = 0. \quad (4.30–4.31)$$

- The coefficient of x_K is $\beta_K + \gamma_2 q$: it can differ across people.
  - **Wage**: $\log(wage) = \cdots + \beta_K educ + \gamma_2 educ \cdot abil + \cdots$: a year of schooling adds $\beta_K + \gamma_2 abil$, more for abler workers if $\gamma_2 > 0$.
- For person i, $\beta_{iK} = \beta_K + \gamma_2 q_i$ is a random coefficient.
  - One slope coefficient is "random"—that is, it depends on individual-specific unobserved heterogeneity.

Partial effect and average partial effect

The effect differs across people; the APE averages those effects

MODEL

$$y = \beta_0 + \beta_1 x_1 + \cdots + \beta_K x_K + \gamma_1 q + \gamma_2 x_K q + v, \quad E(v \mid \mathbf{x}, q) = 0$$

$$\frac{\partial E(y \mid \mathbf{x}, q)}{\partial x_K} = \beta_K + \gamma_2 q. \tag{4.32}$$

The individual effect cannot be recovered from observables without further information or assumptions about q.

$$\text{APE} = E(\beta_K + \gamma_2 q) = \beta_K + \gamma_2 E(q) = \beta_K \quad \text{if } E(q) = 0. \tag{4.33}$$

▸ When the mean of q is not zero ▸ Attendance effects can vary with prior GPA

Average treatment effect

For a binary treatment, the average effect is called the ATE

MODEL

$$y = \beta_0 + \beta_1 x_1 + \cdots + \beta_K x_K + \gamma_1 q + \gamma_2 x_K q + v, \quad E(q) = 0$$

Let $x_K = 1$ for receiving a treatment, such as a job training program or a school voucher, and 0 otherwise.

$$E(y \mid x_1, \ldots, x_{K-1}, 1, q) - E(y \mid x_1, \ldots, x_{K-1}, 0, q) = \beta_K + \gamma_2 q.$$

- The effect of the treatment differs across people with different q.
- Its average over the distribution of q is β_K: the average treatment effect (ATE).

Chapter 19 of the book returns to treatment effects.

What does the schooling coefficient mean?

The main effect is evaluated where the interacting variable equals zero

IQ IS THE PROXY FOR ABILITY

$$\log(wage) = a_0 + \cdots + b_1 educ + b_2 IQ + b_3 educ \cdot IQ + \varepsilon.$$

$$\text{Schooling effect at IQ} = b_1 + b_3 IQ.$$

The coefficient b_1 alone is not the schooling effect at every IQ.

▸ Proxy assumptions for the average effect

At what IQ does the schooling effect equal b_1? How could we move that reference to IQ 100?

Centring changes the reference person

The schooling coefficient now describes someone with average IQ

$IQ_0 = IQ - 100, \qquad c_1 = b_1 + 100b_3.$

	Raw IQ	Centred IQ: $IQ - 100$
Schooling coefficient	.018	.052
Reference person	IQ 0	IQ 100
Interaction coefficient	.00034	.00034

$$.018 + 100 \times .00034 \approx .052.$$

Same fitted wages; a new reference person.

▸ Population versus sample-mean centring

Reading the centred schooling coefficient

Under the proxy assumptions, the centred slope estimates the average partial effect

EXAMPLE 4.5

$$\widehat{\log(wage)} = \cdots + .052\,educ - .00094\,IQ_0 + .00034\,educ \cdot IQ_0, \qquad IQ_0 = IQ - 100.$$

- At IQ 100, one more year of schooling is associated with about 5.2% higher wages.
- If the proxy assumptions hold and $E(IQ) = 100$, this estimates the average partial effect of schooling.

What changes if we observe the variable itself, but measure it inaccurately?

Roadmap

Wooldridge 4.4 — current location in blue

OLS Uncertainty and Inference
- OLS.3 and Standard Errors
- Testing Hypotheses

OLS Solutions to the Omitted Variables Problem
- OLS Ignoring the Omitted Variables
- Proxy Variable–OLS Solution
- Interactions in Unobservables: Random Coefficients

OLS under Measurement Error
- Measurement Error in the Dependent Variable
- Measurement Error in an Explanatory Variable

Another source of endogeneity

The model's variable is well defined, but our data record it with error

LECTURE 3A: THREE SOURCES OF ENDOGENEITY

Omitted variables · Measurement error · Simultaneity

- We have treated **omitted variables**. Next: **measurement error**.
- The model's variable has a clear meaning, such as actual annual income; our data contain a measure of it with error.
- Two cases, each with its own consequence for OLS:
  - error in the dependent variable y: harmless for consistency unless related to the regressors;
  - error in an explanatory variable: classical error pulls the slope toward zero.

Measurement error versus a proxy

A similar statistical structure, but conceptually very different problems

Proxy variable	**Measurement error**
We look for a variable that is somehow associated with the unobserved one.	The unobserved variable has a well-defined, quantitative meaning; our measure of it contains error.
IQ score is a proxy for ability.	Reported annual income is a measure of actual annual income.
The effect of the omitted variable itself cannot be estimated.	The mismeasured variable is often the one whose effect we want.

Roadmap

Wooldridge 4.4.1 — current location in blue

OLS Uncertainty and Inference
- OLS.3 and Standard Errors
- Testing Hypotheses

OLS Solutions to the Omitted Variables Problem
- OLS Ignoring the Omitted Variables
- Proxy Variable–OLS Solution
- Interactions in Unobservables: Random Coefficients

OLS under Measurement Error
- **Measurement Error in the Dependent Variable**
- Measurement Error in an Explanatory Variable

Measurement error in the outcome

We observe the true outcome plus a measurement error

Maintain OLS.1–OLS.2 for the true model:

$$y^* = \beta_0 + \beta_1 x_1 + \cdots + \beta_K x_K + v, \qquad e_0 = y - y^*. \tag{4.37–4.38}$$

What matters is how the measurement error in the population is related to other factors.

$$y = \beta_0 + \beta_1 x_1 + \cdots + \beta_K x_K + \underbrace{(v + e_0)}_{\text{new error}}. \tag{4.39}$$

When does OLS with y in place of y^* produce consistent estimators of the β_j?

When outcome error leaves slopes consistent

Reporting error must be uncorrelated with the regressors

OBSERVED EQUATION

$$y = \beta_0 + \beta_1 x_1 + \cdots + \beta_K x_K + (v + e_0)$$

- The true-model error v is already uncorrelated with the regressors.
- If $\text{Cov}(x_j, e_0) = 0$ for every j, the slope estimators remain consistent.
- A nonzero average reporting error shifts the intercept, not these slopes.

What if treated firms systematically underreport a bad outcome?

Example 4.7: reported scrap rates

Underreporting by grant recipients can make the programme look better

- **Question**: does a job-training grant reduce defective output?
- $scrap^*$: true scrap rate (defective output); lower is better. $grant = 1$ for a grant recipient.

$$\log(scrap^*) = \beta_0 + \beta_1 grant + v, \qquad \log(scrap) = \log(scrap^*) + e_0.$$

- Suppose recipients underreport more: $\text{Cov}(grant, e_0) < 0$.
- The grant coefficient is pushed downward: the measured reduction in scrap looks too large.

▸ Another example: reported saving

Outcome error: what changes the slope?

Systematic reporting can look like an effect of the regressor

ONE-REGRESSOR CASE

$$\operatorname{plim} \hat{\beta}_1 = \beta_1 + \frac{\operatorname{Cov}(x_1, e_0)}{\operatorname{Var}(x_1)}.$$

- If reporting errors tend to be more positive at high x_1, the estimated slope is pushed up; more negative, down.
- If reporting error is uncorrelated with the regressors, the slopes remain consistent.

If outcome noise leaves the slope consistent, does it still have a cost?

Outcome noise can reduce precision

The slope remains consistent, but the estimate becomes less precise

For independent added noise,

$$\text{Var}(v + e_0) = \text{Var}(v) + \text{Var}(e_0).$$

	Recorded log wage	With independent noise added
Schooling slope	.060	.057
Standard error	.006	.008

The estimate changes little; its standard error rises.

Would noise in schooling itself have the same consequence?

Roadmap

Wooldridge 4.4.2 — current location in blue

OLS Uncertainty and Inference
- OLS.3 and Standard Errors
- Testing Hypotheses

OLS Solutions to the Omitted Variables Problem
- OLS Ignoring the Omitted Variables
- Proxy Variable–OLS Solution
- Interactions in Unobservables: Random Coefficients

OLS under Measurement Error
- Measurement Error in the Dependent Variable
- **Measurement Error in an Explanatory Variable**

Measurement error in a regressor

Replacing the true input puts measurement error into the regression error

Maintain OLS.1–OLS.2 for the true model:

$$y = \beta_0 + \beta_1 x_1 + \cdots + \beta_K x_K^* + v. \tag{4.41}$$

- We observe $x_K = x_K^* + e_K$, with $E(e_K) = 0$. (4.42)
- Substitute $x_K^* = x_K - e_K$:

$$y = \beta_0 + \beta_1 x_1 + \cdots + \beta_K x_K + \underbrace{(v - \beta_K e_K)}_{\text{new error}}. \tag{4.44}$$

The observed regressor x_K and the new error both contain e_K. Under the maintained error assumptions, $\mathrm{Cov}(x_K, e_K) = 0$ (4.43) preserves consistency; details in the appendix. ▸ Error uncorrelated with the recorded value

Classical error creates endogeneity

The observed regressor and the new error share the same recording noise

$x_K = x_K^* + e_K, \qquad u = v - \beta_K e_K.$

Classical error: noise is uncorrelated with the truth,

$$\text{Cov}(x_K^*, e_K) = 0. \tag{4.45}$$

Maintain $E(e_K) = 0$, $\text{Cov}(v, x_K^*) = \text{Cov}(v, e_K) = 0$ and $\text{Cov}(e_K, x_j) = 0$ for $j < K$.

$$\text{Cov}(x_K, u) = -\beta_K \,\text{Var}(e_K).$$

If $\beta_K \neq 0$, OLS.1 fails.

What happens to the fitted slope when the input moves without the outcome?

Classical error attenuates a single slope

The slope is multiplied by the share of variation coming from the true input

ONE REGRESSOR

$$y = \beta_0 + \beta_1 x_1 + (v - \beta_1 e_1), \quad x_1 = x_1^* + e_1$$

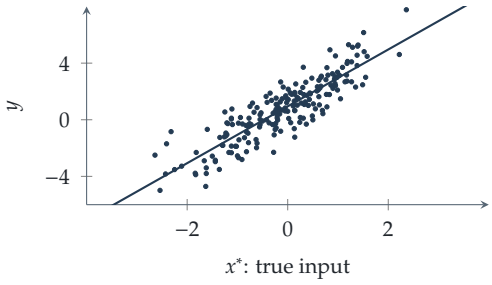
$$\operatorname{plim} \hat{\beta}_1 = \beta_1 \underbrace{\frac{\operatorname{Var}(x_1^*)}{\operatorname{Var}(x_1^*) + \operatorname{Var}(e_1)}}_{\text{true-input share of recorded variation}}. \tag{4.48}$$

- Between 0 and 1: slopes move toward zero; more data do not help.

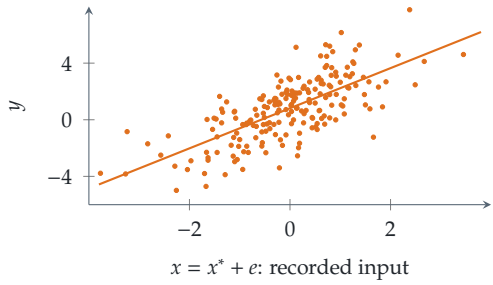
▸ Deriving the attenuation factor ▸ Attenuation with several regressors

Measurement error flattens the fitted line

Same observations and outcomes; extra movement only in the measured input



Fitted slope: 2.01



Fitted slope: 1.42

More horizontal variation, with the same outcomes: a flatter fitted line.

‣ Responses to measurement error

Example 4.8: reported family income

Attenuation makes a real effect harder to detect

$$colGPA = \beta_0 + \beta_1 faminc^* + \beta_2 hsGPA + \beta_3 SAT + v, \qquad faminc = faminc^* + e_1.$$

- College GPA, high-school GPA and SAT come from school records; students report family income.
- Under classical error, the income coefficient is biased toward zero.
- A test of $H_0 : \beta_1 = 0$ then has a higher probability of Type II error: a real effect can look insignificant.

When classical measurement error is doubtful

The attenuation result rests on the classical assumption

CLASSICAL ERROR

$$x_K = x_K^* + e_K, \quad \text{Cov}(x_K^*, e_K) = 0$$

- **Days of marijuana use**: nonsmokers report 0 exactly, while smokers miscount, and reports cannot fall below 0 or exceed 30. The error is then very likely correlated with the true value.

Measurement error: what to take into practice

Error in an explanatory variable is usually the more serious problem

1. **Error in y**: if it is uncorrelated with the regressors, OLS slopes remain consistent. Independent noise can make estimates less precise.
2. **Error in x**: under the classical assumptions, the mismeasured regressor's slope is attenuated toward zero. More observations do not remove this distortion.
3. **Before interpreting a small coefficient**: check how the variable was recorded. A weak estimate may reflect noisy data; outside the classical case, the direction is not guaranteed.

▸ Practice: which way does the bias go?

What should we check before calling a regression coefficient an effect?

Wrap-up (3b)

What to take into empirical work

Read the coefficient, assess uncertainty, then examine what could distort it

1. **OLS and inference**: define the target; use standard errors, t and F to assess sampling uncertainty. (4.2.1–4.2.3)
2. **Omitted variables**: name the missing factor and sign its two links before giving a causal interpretation. (4.3.1)
3. **Proxies**: explain why the proxy removes the remaining regressor–error link; adding a control alone is not enough. (4.3.2)
4. **Interactions**: centre at the relevant mean to read an average effect, under the identifying assumptions. (4.3.3)
5. **Measurement error**: unrelated outcome noise can cost precision; classical regressor noise attenuates the slope. (4.4)

▸ Reading map and assumption checklist

Next lecture

No lecture on the holiday Monday; then instrumental variables

- Today: Linear Regression, 3b — Wooldridge (2010), Chapter 4, Sections 4.2.2–4.4.
- Next: Instrumental Variables, Wooldridge (2010), Chapter 5 (Weeks 6–7): what to do when a regressor stays correlated with the error.
- No lecture on Monday 5 October 2026 (National Day holiday); Instrumental Variables begins the following Monday.
- Problem Set 1 due Sunday 11 October 2026. Will be posted this week.

▸ Appendix contents

Adapted from slides by Shuyang Sheng and Qihui Chen

Appendix

Appendix contents

Optional explanations with returns to the relevant concept

C Lagrange multiplier tests App. 1

D Inference: derivations and test details App. 7

E OVB, selection and proxies App. 30

F Measurement error: derivations and practice App. 58

◂ Back to next lecture and PS1

Appendix roadmap

APPENDIX C · Wooldridge 4.2.4 — current location in blue

Lagrange Multiplier Tests

Inference Illustrations

OVB, Selection and Proxies

Measurement Error

Back to contents

The LM (score) statistic

estimate under H_0, then ask whether the excluded regressors explain the residuals

- Partition $y = \mathbf{x}_1\boldsymbol{\beta}_1 + \mathbf{x}_2\boldsymbol{\beta}_2 + u$ and test $H_0 : \boldsymbol{\beta}_2 = \mathbf{0}$ (K_2 restrictions).
- Estimate the *restricted* model (y on $\mathbf{x}_1$) and keep its residuals $\tilde{u}_i$. Under H_0, $\mathbf{x}_{i2}$ should be uncorrelated with $\tilde{u}_i$ up to sampling error.
- Regress $\tilde{u}$ on *all* of $\mathbf{x}_1, \mathbf{x}_2$ (including $\mathbf{x}_1$ is essential) and compute $LM = NR_u^2$. Under H_0 and OLS.3, $LM \overset{a}{\sim} \chi^2_{K_2}$. Also called the N-R^2 test.
- **Example 4.1** (MROZ, H_0: no age or children effects): $R_u^2 \approx .0017$; from unrounded R_u^2, $LM \approx .722$, p-value $\approx .87$ — the same verdict as $F = .24$ ($p \approx .87$).

◂ Back to the Wald rejection rule ◂ Back to the F statistic ▸ Robust LM

Heteroskedasticity-robust LM

three auxiliary regressions and $N - SSR_0$

- Regress each element of $\mathbf{x}_2$ on $\mathbf{x}_1$ and keep the residuals $\hat{r}_{i1}, \ldots, \hat{r}_{iK_2}$ (the part of $\mathbf{x}_2$ not explained by $\mathbf{x}_1$).
- Regress the constant 1 on $\tilde{u}_i\hat{r}_{i1}, \ldots, \tilde{u}_i\hat{r}_{iK_2}$ without an intercept; with SSR_0 its sum of squared residuals, $LM = N - SSR_0 \overset{a}{\sim} \chi^2_{K_2}$ under H_0 whatever the form of heteroskedasticity.
- **Example 4.1** continued: $LM = .51$, $p \approx .92$; the robust Wald statistic gives $p \approx .92$. Tiny statistics either way: these tests do not establish an additional linear association with age or children in this sample.

‹ Back to the Wald rejection rule ‹ Back to the F statistic ‹ Back to the LM statistic

Appendix roadmap

APPENDIX D · Wooldridge 4.2.2–4.2.3 — current location in blue

Lagrange Multiplier Tests

Inference Illustrations

OVB, Selection and Proxies

Measurement Error

Back to contents

Homoskedasticity implies OLS.3

Constant conditional variance is sufficient, not necessary

Assumed: zero conditional mean and constant conditional variance,

$$E(u \mid \mathbf{x}) = 0, \qquad \text{Var}(u \mid \mathbf{x}) = \sigma^2.$$

Implied: $E(u^2 \mid \mathbf{x}) = \sigma^2$, and then, by iterated expectations,

$$E(u^2\mathbf{x}'\mathbf{x}) = \sigma^2 E(\mathbf{x}'\mathbf{x}) \qquad \text{(OLS.3)}.$$

- The reverse can fail: OLS.3 is the weaker condition.
- **Homoskedasticity**: $\text{Var}(u \mid \mathbf{x}) = \sigma^2$.

‣ Why constant variance implies OLS.3 ◂ Back to the homoskedasticity recap
‣ OLS.3 in the textbook

Assumption OLS.3

The weakest form of homoskedasticity

Assumption OLS.3

$$E(u^2\mathbf{x}'\mathbf{x}) = \sigma^2 E(\mathbf{x}'\mathbf{x}), \qquad \sigma^2 = E(u^2).$$

Element by element: u^2 is uncorrelated with every x_j, every x_j^2 and every product $x_j x_k$,

$$\text{Cov}(u^2, x_j x_k) = 0 \quad \text{for all } j, k.$$

◂ Back to the homoskedasticity recap ▸ Variance and standard error
◂ Back to the OLS.3 implication

What OLS.3 buys

A simple variance with three ingredients

Under OLS.1–OLS.3 (Theorem 4.2), with $\mathbf{A} = E(\mathbf{x}'\mathbf{x})$,

$$\text{Avar}(\hat{\boldsymbol{\beta}}) = \frac{\sigma^2}{N}\,\mathbf{A}^{-1}.$$

One coefficient (R_j^2: share of x_j explained by the other regressors):

$$\text{Avar}(\hat{\beta}_j) = \frac{\sigma^2}{N\ \text{Var}(x_j)\,(1 - R_j^2)}.$$

Next: estimate numbers on σ^2 and $\mathbf{A}$ from one sample, and read a standard error.

‹ Back to the homoskedasticity recap · › Estimating the standard error
‹ Back to Assumption OLS.3

From the variance to a standard error

Estimate the two unknowns from our one sample

Avar($\hat{\boldsymbol{\beta}}$) = $\sigma^2[E(\mathbf{x}'\mathbf{x})]^{-1}/N$ under OLS.1–OLS.3.

- From the sample (K counts the intercept):

$$\hat{\sigma}^2 = \frac{SSR}{N-K'}, \qquad \frac{\mathbf{X}'\mathbf{X}}{N} \quad \Longrightarrow \quad \hat{\mathbf{V}} = \hat{\sigma}^2(\mathbf{X}'\mathbf{X})^{-1}. \tag{4.10}$$

- **Standard error**: se($\hat{\beta}_j$) = $\sqrt{\hat{V}_{jj}}$, the number software prints beside each coefficient.
- If OLS.3 fails, use robust standard errors; $\hat{\boldsymbol{\beta}}$ is unchanged.

▸ Robust standard errors ◂ Back to the homoskedasticity recap
◂ Back to the variance formula

Why constant variance implies OLS.3

Two steps: a second-moment identity, then iterated expectations

ASSUME

$E(u \mid \mathbf{x}) = 0, \qquad \mathrm{Var}(u \mid \mathbf{x}) = \sigma^2$

$$E(u^2 \mid \mathbf{x}) = \mathrm{Var}(u \mid \mathbf{x}) + [E(u \mid \mathbf{x})]^2 = \sigma^2.$$

By iterated expectations, for every pair j, k,

$$E(u^2 x_j x_k) = E\Big[x_j x_k \, E(u^2 \mid \mathbf{x})\Big] = \sigma^2 E(x_j x_k).$$

Stacked over all pairs, this is OLS.3. OLS.3 need not imply either conditional restriction.

◂ Back to homoskedasticity and OLS.3 ◂ Back to the OLS.3 implication

Robust standard errors

Allow unequal error variances; keep the OLS coefficients

Without OLS.3: $\text{Avar}(\hat{\boldsymbol{\beta}}) = \mathbf{A}^{-1}\mathbf{B}\mathbf{A}^{-1}/N$, with $\mathbf{B} = E(u^2\mathbf{x}'\mathbf{x})$.

$$\hat{\mathbf{V}}_{\text{robust}} = (\mathbf{X}'\mathbf{X})^{-1}\Big(\sum_{i=1}^{N} \hat{u}_i^2\mathbf{x}_i'\mathbf{x}_i\Big)(\mathbf{X}'\mathbf{X})^{-1}. \tag{4.11}$$

- Each squared residual enters the uncertainty calculation separately.
- Diagonal square roots give heteroskedasticity-robust standard errors.
- With OLS.1–OLS.2 and the usual regularity conditions, these remain asymptotically valid when OLS.3 fails.

Robust variance: derivation and adjustment · The robust column in the wage example · Back to standard errors · Back to estimating the standard error

Estimating the robust variance

Replace the unknown errors by OLS residuals

$$\text{Avar}(\hat{\boldsymbol{\beta}}) = \frac{\mathbf{A}^{-1}\mathbf{B}\mathbf{A}^{-1}}{N}, \qquad \mathbf{B} = E(u^2\mathbf{x}'\mathbf{x}).$$

- Estimate $\mathbf{A}$ by $\mathbf{X}'\mathbf{X}/N$ and $\mathbf{B}$ by $\hat{\mathbf{B}} = N^{-1}\sum_i \hat{u}_i^2\mathbf{x}_i'\mathbf{x}_i$.
- Substitution gives (4.11). The **HC1 adjustment** multiplies that variance matrix by $N/(N-K)$.

Small-sample performance and weighted least squares | Back to robust standard errors
Back to standard errors

Reading the robust column

Same coefficient; a different uncertainty estimate

$$\hat{\mathbf{V}}_{\text{HC1}} = \frac{N}{N-K}(\mathbf{X}'\mathbf{X})^{-1}\Big(\sum_{i=1}^{N}\hat{u}_i^2\mathbf{x}_i'\mathbf{x}_i\Big)(\mathbf{X}'\mathbf{X})^{-1}.$$

- Robust S.E. = $\sqrt{\text{corresponding diagonal element}}$.
- Example 4.1, *kidslt6*: $\hat{\beta}_5 = -.061$ either way; the usual S.E. is .089 and the robust S.E. is .106.

‹ Back to robust standard errors ‹ Back to standard errors

Robust standard errors: costs; why not WLS?

robust is free asymptotically, not in small samples

- (4.11) is consistent under *any* heteroskedasticity, including none. In small samples it can be less accurate than the usual estimator when OLS.3 holds; the degrees-of-freedom correction $N/(N-K)$ (MacKinnon and White 1985) helps a little, and software (HC1) uses it.
- Weighted least squares needs a correct model of $\text{Var}(y \mid \mathbf{x})$. More subtly, WLS is generally inconsistent for the linear projection when $E(u \mid \mathbf{x}) \neq 0$ but OLS.1 holds — so it is the wrong tool for a projection.
- With large N, heteroskedasticity rarely threatens inference by OLS; the point is only to compute the standard errors appropriately.

Back to the robust variance derivation
Back to robust standard errors
Back to standard errors

Two standard errors on the ATTEND fit

the same slope, the usual and the robust se

$$\widehat{stndfnl} = -.634 + .00813\, atndrte, \qquad N = 680, \quad R^2 = .0196$$

	slope $\hat{\beta}_1$	usual se	robust se (HC1)
atndrte	.00813	.00221	.00212
$t = \hat{\beta}_1 / \text{se}$		3.68	3.84

Both standard errors lead to rejection of a zero slope at the 5% level.

‹ Back to standard errors (4.2.2)

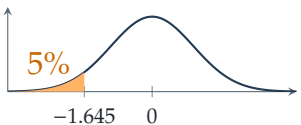
One-sided alternatives

When theory says which sign to expect

- $H_0 : \beta_j = 0$ against $H_1 : \beta_j > 0$ (or $H_1 : \beta_j < 0$).
- Reject in favour of $\beta_j > 0$ if $\hat{t}_j > c$, where c is the 95th percentile: all 5% in one tail ($c = 1.645$ for large $N - K$).
- A negative $\hat{t}_j$, however large, never rejects in favour of $\beta_j > 0$.



$H_1 : \beta_j > 0$



$H_1 : \beta_j < 0$

Back to the null hypothesis

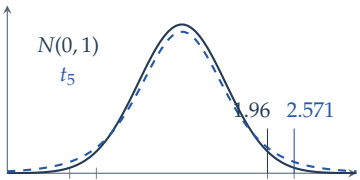
The t distribution

Like the standard normal, with heavier tails when degrees of freedom are few

Definition: if $Z \sim N(0,1)$ and $X \sim \chi^2_\nu$ are independent,

$$T = \frac{Z}{\sqrt{X/\nu}} \sim t_\nu.$$

- Symmetric around zero; heavier tails than $N(0,1)$.
- As ν grows, t_ν approaches $N(0,1)$.
- With normal errors, $\hat{t}_j \sim t_{N-K}$.



Back to the t statistic

Testing a value other than zero

Distance from the hypothesized value, in standard errors

$$H_0 : \beta_j = a_j : \qquad \hat{t}_j = \frac{\hat{\beta}_j - a_j}{\text{se}(\hat{\beta}_j)} = \frac{\text{estimate} - \text{hypothesized value}}{\text{standard error}}.$$

- It counts how many standard errors $\hat{\beta}_j$ lies from a_j; with $a_j = 0$ it is the usual $\hat{t}_j$.
- Example 4.1, is the return to schooling 10%? $H_0 : \beta_3 = .10$: $\hat{t}_3 = (.108 - .10)/.014 \approx .57 < 1.96$, fail to reject.

Back to the t statistic

Critical values of the t distribution

Two-sided, by significance level and degrees of freedom

Degrees of freedom	$\alpha = 10\%$	$\alpha = 5\%$	$\alpha = 1\%$
5	2.015	2.571	4.032
10	1.812	2.228	3.169
20	1.725	2.086	2.845
30	1.697	2.042	2.750
60	1.671	2.000	2.660
120	1.658	1.980	2.617
∞ ($N(0,1)$)	1.645	1.960	2.576

- Smaller α, larger c; more degrees of freedom, smaller c. With $N - K > 120$, use the $N(0,1)$ row.

Back to Type I error

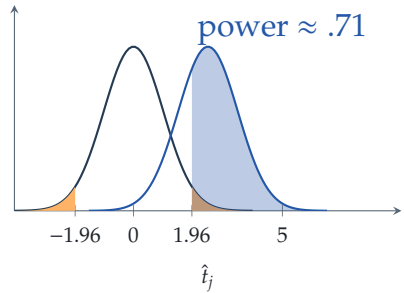
Sample size and power

More observations can make a fixed alternative easier to detect

- A **consistent test** satisfies, for a fixed alternative,

$$\lim_{N\to\infty} \Pr_N(\text{reject } H_0 \mid H_1) = 1.$$

- Suppose $\beta_j / \operatorname{se}(\hat{\beta}_j) \approx 2.5$ at sample size N.



Illustrative normal approximation. Orange: rejection under the null; blue: rejection under the specified alternative.

‹ Back to Type II error and power

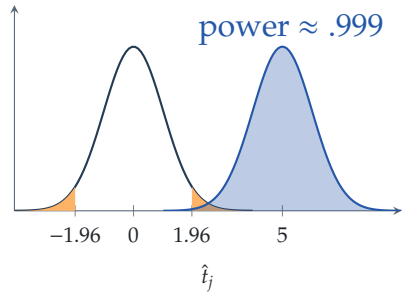
Sample size and power

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$$\lim_{N\to\infty} \Pr_N(\text{reject } H_0 \mid H_1) = 1.$$

- Suppose $\beta_j / \operatorname{se}(\hat{\beta}_j) \approx 2.5$ at sample size N.
- At $4N$, the S.E. approximately halves; the signal becomes 5 S.E.s.



Illustrative normal approximation. Orange: rejection under the null; blue: rejection under the specified alternative.

‹ Back to Type II error and power

Two ways to run a t test

A critical value or a p-value gives the same decision

ONE COEFFICIENT

$H_0 : \beta_j = 0, \qquad \hat{t}_j = \hat{\beta}_j / \text{se}(\hat{\beta}_j)$

Critical value

1. Choose α, say 5%.
2. Find c: the 97.5th percentile (≈ 1.96).
3. Compute $\hat{t}_j$.
4. Reject if $|\hat{t}_j| > c$.

p-value

1. Compute $\hat{t}_j$.
2. Compute $p = \Pr(|T| \geq |\hat{t}_j|)$.
3. Reject if $p < \alpha$.
4. Report p: it shows how strong the evidence is.

Same decision: $p < \alpha$ exactly when $|\hat{t}_j| > c$.

◂ Back to the p-value example

Statistical versus economic significance

Significant is not the same as large; insignificant is not the same as zero

WHAT MOVES THE STANDARD ERROR

$\text{Avar}(\hat{\beta}_j) = \sigma^2/[N\ \text{Var}(x_j)\,(1 - R_j^2)]$: noise, sample size, own variation of x_j.

1. **Significant**: now judge the size (units, logs). With a large N, tiny effects become significant.
2. **Not significant**: is the estimate still large? Report the p-value; a large estimate with a large p is imprecise, not zero. The young-children estimate: about -6% per child, $p = .49$.
3. **Unexpected sign**: check uncertainty, coding and the model; consider omitted variables.

‹ Back to the young-children interval

What intervals and tests assume

An interval is only as good as the assumptions behind it

- **Omitted factors** correlated with the regressors: OLS is inconsistent, so the interval is centred in the wrong place. The next section.
- **Heteroskedasticity**: the usual standard error is not valid, so the interval is not truly 95%. Use robust standard errors.
- **Normal errors**: not important with hundreds of observations; the large-sample route gives the same intervals.

‹ Back to confidence intervals

Other wage coefficients

Apply the same test coefficient by coefficient

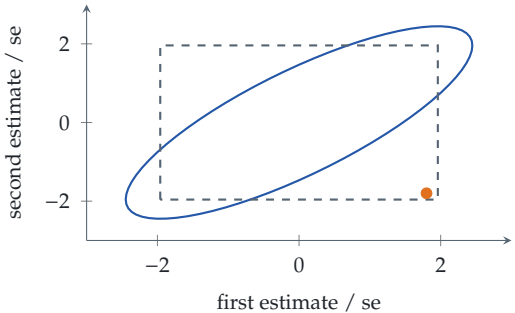
	Usual		Robust	
Coefficient	t	p	t	p
educ	7.49	$< .001$	7.92	$< .001$
exper	2.97	.003	2.61	.009

- Reject each corresponding zero-coefficient null at 5%, using either variance estimate.

Back to the young-children interval

Separate tests and a joint test

a joint test uses covariance between estimates



- Illustrative $H_0 : \beta_1 = \beta_2 = 0$; estimates $(1.8, -1.8)$, both se = 1, estimated correlation .8.
- Separate tests: both $|t| = 1.8 < 1.96$.
- Joint F test: $F = 16.2 > 3.00$; reject at 5%.

Blue ellipse: joint 5% nonrejection region; dashed square: both individual 5% tests fail to reject.

◂ Back to several restrictions at once (4.2.2)

The Wald statistic

The large-sample version of the F test

EXCLUSION HYPOTHESIS

$y = \mathbf{x}_1\boldsymbol{\beta}_1 + \mathbf{x}_2\boldsymbol{\beta}_2 + u, \qquad H_0 : \boldsymbol{\beta}_2 = \mathbf{0}$

$$W = \hat{\boldsymbol{\beta}}_2' \hat{\mathbf{V}}_{22}^{-1} \hat{\boldsymbol{\beta}}_2 \overset{a}{\sim} \chi^2_{K_2} \quad \text{under } H_0.$$

- $\hat{\mathbf{V}}_{22}$ estimates the variance matrix of $\hat{\boldsymbol{\beta}}_2$, using the usual or robust formula.
- The matrix accounts for both variances and covariances among the estimates.
- $W = Q \cdot F$ with $Q = K_2$; with one restriction, $W = \hat{t}^2$. A robust $\hat{\mathbf{V}}$ gives a robust joint test.

▸ The Wald rejection rule ▸ General linear restrictions ◂ Back to the F statistic

The Wald rejection rule

Large values are unusual under the joint null

THE WAGE HYPOTHESIS

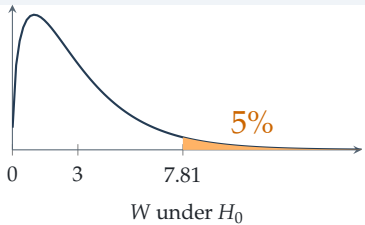
$H_0 : \beta_4 = \beta_5 = \beta_6 = 0, \qquad K_2 = 3, \qquad W \overset{a}{\sim} \chi^2_3$

- At 5%, reject when $W > 7.81$.

$$\Pr(\chi^2_3 > 7.81) \approx .05.$$

- Equivalently, reject when

$$p = \Pr(\chi^2_3 > W) < .05.$$



Example 4.1: usual $W = .711 = 3F$, robust $W = .493$; $p = .87$ and .92. Fail to reject either way.

‣ The LM alternative ‣ Back to the Wald statistic ‣ Back to the F statistic

General linear restrictions

A matrix can collect several coefficient restrictions

$$H_0 : \mathbf{R}\boldsymbol{\beta} = \mathbf{r}. \tag{4.12}$$

- $\boldsymbol{\beta}$ is $K \times 1$; $\mathbf{R}$ is $Q \times K$ with rank $Q \leq K$; $\mathbf{r}$ is $Q \times 1$.
- To test $\boldsymbol{\beta}_2 = \mathbf{0}$,

$$\mathbf{R} = (\mathbf{0} \;\; \mathbf{I}_{K_2}), \qquad \mathbf{r} = \mathbf{0}, \qquad Q = K_2.$$

- Thus $\mathbf{R}\hat{\boldsymbol{\beta}} - \mathbf{r} = \hat{\boldsymbol{\beta}}_2$.

Wald formula for general restrictions | Back to the exclusion Wald statistic | Back to the F statistic

The general Wald statistic

Scale the restriction errors by their covariance matrix

GENERAL LINEAR HYPOTHESIS

$H_0 : \mathbf{R}\boldsymbol{\beta} = \mathbf{r}$, $\quad Q$ restrictions

$$W = (\mathbf{R}\hat{\boldsymbol{\beta}} - \mathbf{r})'(\mathbf{R}\hat{\mathbf{V}}\mathbf{R}')^{-1}(\mathbf{R}\hat{\boldsymbol{\beta}} - \mathbf{r}). \tag{4.13}$$

- Under H_0, $W \overset{a}{\sim} \chi^2_Q$.
- $\hat{\mathbf{V}}$ is the usual or robust estimated variance matrix, according to the maintained assumptions.

◂ Back to general linear restrictions ◂ Back to the exclusion Wald statistic
◂ Back to the F statistic

One restriction: t and F agree

Use t for one coefficient and F for a group

- For one restriction, $F = \hat{t}_j^2$ and t_{N-K}^2 has an $F_{1,N-K}$ distribution: the same two-sided decision.
- The t statistic is more flexible for one coefficient: it also handles one-sided alternatives.
- A joint F can hide one significant variable inside a group of insignificant ones. Report the separate t tests too.

Back to the F statistic

Which test for which hypothesis

The hypothesis determines the statistic and its reference distribution

Hypothesis	Reference under H_0
One coefficient: $\beta_j = a_j$	$\hat{t}_j = (\hat{\beta}_j - a_j)/\,\mathrm{se}(\hat{\beta}_j)$: t_{N-K}, or $N(0,1)$ for large N
An interval for β_j	$\hat{\beta}_j \pm c \cdot \mathrm{se}(\hat{\beta}_j)$
A group: Q exclusions	F: $F_{Q,N-K}$; large-sample $W = Q \cdot F \overset{a}{\sim} \chi^2_Q$

- The usual standard errors require OLS.3; robust ones allow its failure.
- With one restriction, $F = \hat{t}^2$.

‹ Back to the wage joint-test decision

The big picture, backwards

why we needed today for the causal goal, with where to read further

- **Goal**: a causal β_K — the effect of x_K holding everything else, observed or not, fixed (4.1).
- **OLS reaches it** under OLS.1–OLS.2 and the regularity conditions; it is consistent and approximately normal (4.2.1–4.2.2), and a standard error makes the estimate usable (4.2.2–4.2.3, today's first hour).
- **What threatens OLS.1**: an omitted q (4.3.1), a mismeasured x_K (4.4.2), simultaneity (Chapter 9).
- **Today's fixes and their limits**: a proxy that is redundant and sufficient (4.3.2–4.3.3); better measurement (4.4); otherwise an instrument (Chapter 5, next).

‹ Back to the lecture takeaways

Appendix roadmap

APPENDIX E · Wooldridge 4.1, 4.3.1–4.3.3 — current location in blue

Lagrange Multiplier Tests

Inference Illustrations

OVB, Selection and Proxies

Measurement Error

‹ Back to contents

Recall: the structural target

A structural coefficient describes a specified ceteris paribus relationship

$$\log(wage) = \beta_0 + \beta_1 exper + \beta_2 exper^2 + \beta_3 educ + \gamma abil + v.$$

- The target β_3 compares schooling while holding experience and ability fixed.
- Ability remains in this population relationship even when the dataset does not measure it.
- A regression omitting ability estimates a different projection unless the required orthogonality conditions hold.

Back to omission and endogeneity

Why the argument uses probability limits

Orthogonality gives consistency; unbiasedness needs stronger conditions

$$y = (\beta_0 + \gamma\delta_0) + \sum_{j=1}^{K} (\beta_j + \gamma\delta_j)x_j + e,$$

$$e = v + \gamma r, \qquad E(e) = 0, \quad E(x_j e) = 0.$$

- With full rank and random sampling, these OLS.1 moments give consistency by Theorem 4.1.
- They need not imply $E(e \mid \mathbf{x}) = 0$, so finite-sample unbiasedness does not follow.

◂ Back to the omitted-variable probability limit

The quiz question about controls

A control must address the omitted factor under stated assumptions

A QUESTION FROM QUIZ 1

"Can controls really remove self-selection bias if motivation can't be measured? By what approaches?"

- An observed proxy can help if it supplies information about the missing factor.
- Redundancy and sufficiency make the remaining error orthogonal to the regressors.
- Merely adding controls, or observing more workers, does not establish those conditions.

‹ Back to the two proxy requirements ‹ Back to Quiz 1

From redundancy to error orthogonality

The conditional-mean restriction implies the moment used by OLS

$$E(y \mid \mathbf{x}, q, z) = E(y \mid \mathbf{x}, q) = \beta_0 + \mathbf{x}\boldsymbol{\beta} + \gamma q.$$

$$v = y - \beta_0 - \mathbf{x}\boldsymbol{\beta} - \gamma q \quad \Longrightarrow \quad E(v \mid \mathbf{x}, q, z) = 0.$$

$$E(zv) = E\big[zE(v \mid \mathbf{x}, q, z)\big] = 0.$$

The reverse implication does not follow: one zero moment does not fix an entire conditional-mean function.

Back to redundancy

What if the proxy is imperfect?

Remaining ability related to schooling leaves a remaining distortion

$$q = \theta_0 + \rho_1 x_1 + \cdots + \rho_K x_K + \theta_1 z + \eta,$$

where η is orthogonal to $(1, \mathbf{x}, z)$ by projection.

$$\operatorname{plim} \hat{\beta}_{j,\text{proxy}} = \beta_j + \gamma \rho_j.$$

- Sufficiency is the restriction $\rho_1 = \cdots = \rho_K = 0$.
- A proxy can reduce the distortion without eliminating it; improvement is not guaranteed.

‣ How an unsuitable control can worsen bias ‣ Back to the two proxy requirements

An unsuitable control can amplify bias

A variable explaining only the exogenous part of the regressor can hurt

$$y = \beta_0 + \beta_1 x + u, \qquad \text{Cov}(x, u) \neq 0.$$

- Consider adding a with $\text{Cov}(a, u) = 0$ but $\text{Cov}(a, x) \neq 0$.
- Partialling out a removes variation from x without removing its covariance with u.
- The remaining variation in x contains a larger share of the endogeneity.

‣ The scalar bias calculation ‣ Back to imperfect proxies ‣ Main: proxy requirements

The scalar bias-amplification calculation

The covariance is unchanged and the denominator falls

$$s = x - L(x \mid 1, a), \qquad \operatorname{Cov}(a, u) = 0.$$

$$\operatorname{Cov}(s, u) = \operatorname{Cov}(x, u), \qquad 0 < \operatorname{Var}(s) < \operatorname{Var}(x).$$

$$\operatorname{plim} \hat{\beta}_{\text{omit } a} - \beta_1 = \frac{\operatorname{Cov}(x, u)}{\operatorname{Var}(x)},$$

$$\operatorname{plim} \hat{\beta}_{\text{include } a} - \beta_1 = \frac{\operatorname{Cov}(x, u)}{\operatorname{Var}(s)}.$$

The absolute inconsistency increases when the numerator is nonzero.

‹ Back to unsuitable controls › Main: proxy requirements

Other readings of the IQ table

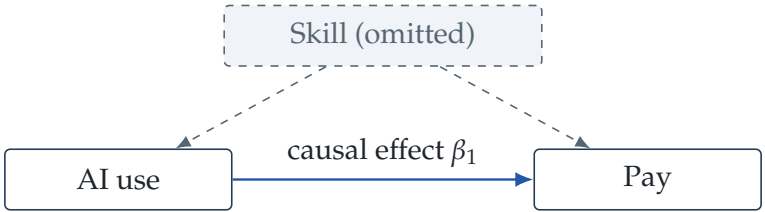
Fit, precision and identification answer different questions

- R^2 rises from .253 to .263; fit improves only slightly.
- The schooling coefficient falls from .065 to .054; a small change in fit can accompany a meaningful change in a coefficient.
- Its standard error rises from .006 to .007: adding a proxy need not improve every coefficient's precision.
- Neither the change in fit nor the change in coefficient proves proxy validity.

The schooling estimates with intervals · Back to the schooling-coefficient decline

Return to the AI-and-pay quiz question

Self-selection can make AI users different before adoption



Reported comparison: AI users earn about 15% more.

Suppose workers who use AI also have higher skill, and skill raises pay.

How much of the reported pay gap might be the skill difference?

‣ Sign the two links ‣ Original quiz question ‣ Back to the two links in omitted ability

Why the pay gap can overstate the AI effect

The same two-link omitted-variable argument applies

$$y = \log(pay) = \beta_0 + \beta_1 D + \gamma q + v; \quad q = \delta_0 + \delta_1 D + r$$

$$\operatorname{plim} \hat{\beta}_1 = \beta_1 + \gamma\delta_1.$$

1. $\gamma > 0$: greater skill raises pay, holding AI use fixed.
2. $\delta_1 > 0$: AI users have higher average skill than nonusers.

The positive skill contribution makes the pay comparison overstate the AI effect.

Group-gap derivation | Back to the AI diagram | Return to OVB

Revisiting quiz 1: the AI-and-pay headline

A pay comparison leaves a causal question open

THE SETUP

Workers who use generative-AI tools earn about 15% more than similar workers who do not (hypothetical). Headline: “Using AI raises your pay by 15%.”

- The headline compares AI users with non-users.
- Would the same worker earn more because of using AI?

‹ Back to the AI-and-pay diagram › Return to omitted ability
› Quiz question about controls

The AI comparison and OVB

For a binary regressor, the auxiliary slope is a group-mean difference

WITHIN THE SAME OBSERVED CHARACTERISTICS

$$y = \beta_0 + \beta_1 D + \gamma q + v, \quad E(v \mid D, q) = 0$$

$$q = \delta_0 + \delta_1 D + r, \qquad D \in \{0, 1\}.$$

$$\delta_1 = \frac{\mathrm{Cov}(D, q)}{\mathrm{Var}(D)} = E(q \mid D = 1) - E(q \mid D = 0).$$

$$\operatorname{plim} \hat{\beta}_{1,\text{omit}} - \beta_1 = \gamma \, [E(q \mid D = 1) - E(q \mid D = 0)].$$

Positive sorting on a wage-raising skill gives a positive selection contribution.

‹ Back to the AI two-link explanation › Return to omitted ability

Checking the proxy regression's error

Both parts of the new error must be orthogonal to every regressor

$$e = \gamma r + v, \qquad abil = \theta_0 + \theta_1 IQ + r.$$

$$E(e) = \gamma E(r) + E(v) = 0,$$

$$E(x_j e) = \gamma E(x_j r) + E(x_j v) = 0,$$

$$E(IQ \cdot e) = \gamma E(IQ \cdot r) + E(IQ \cdot v) = 0.$$

- $E(x_j r) = 0$: sufficiency; $E(IQ \cdot r) = 0$: projection.
- $E(x_j v) = 0$: structural conditional mean; $E(IQ \cdot v) = 0$: redundancy.

◂ Back to the assumptions that make IQ useful

Two proxies

More proxies still require redundancy and sufficiency

$$E(y \mid \mathbf{x}, q, z_1, z_2) = E(y \mid \mathbf{x}, q).$$
$$L(q \mid 1, \mathbf{x}, z_1, z_2) = L(q \mid 1, z_1, z_2).$$

- Under these two conditions, include both proxies in the additive regression.
- NLS80 also records KWW, a knowledge-of-work score (Problem 4.11).

Back to the assumptions that make IQ useful | Back to the equation map

How the equations fit together

The same structural target; two different ways to replace the unobserved factor

$$y = \beta_0 + \sum_j \beta_j x_j + \gamma q + v.$$

1. **Omit** q: project it on the observed regressors.

$$q = \delta_0 + \sum_j \delta_j x_j + r \quad \Longrightarrow \quad \text{slope: } \beta_j + \gamma\delta_j.$$

2. **Use a proxy** z: project q on z and impose the two proxy assumptions.

$$q = \theta_0 + \theta_1 z + r \quad \Longrightarrow \quad \text{slope: } \beta_j; \quad \text{error: } \gamma r + v.$$

The two r terms are different residuals. δ_j links q to x_j; θ_1 links q to the proxy.

‹ Back to the proxy setup › Two proxies

Why the proxy regression works

The new error satisfies the orthogonality condition for OLS

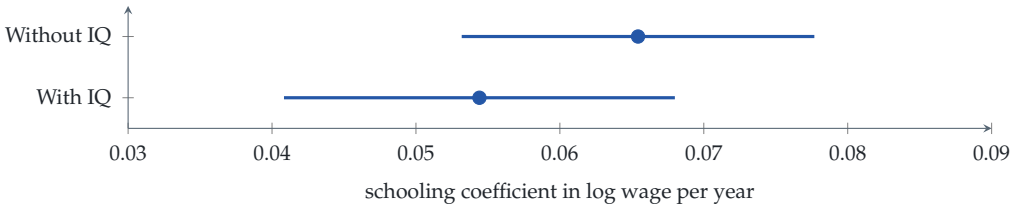
$$y = (\beta_0 + \gamma\theta_0) + \beta_1 x_1 + \cdots + \beta_K x_K + \gamma\theta_1 z + \underbrace{(\gamma r + v)}_{\text{new error}}. \tag{4.28}$$

- **Sufficiency**: r is uncorrelated with every x_j, as v already is. **Redundancy**: z is uncorrelated with v; projection makes it uncorrelated with r.
- Thus OLS.1 holds for $(1, \mathbf{x}, z)$. With OLS.2, Theorem 4.1 gives consistent estimates of the original slopes β_j.

Back to the assumptions that make IQ useful

Adding IQ changes the schooling coefficient

compare estimates and uncertainty on the same sample



- Same 935 men and observed controls; only the IQ regressor changes.
- The schooling coefficient falls from 0.0654 to 0.0544. This sample comparison does not verify the proxy assumptions.

Bars: coefficient ±1.96 se, usual standard errors. Overlap is not a test.

‹ Back to other readings of the IQ table › Main: schooling decline

Example 4.4: a lagged outcome as a proxy

last year's scrap rate stands in for unobserved firm productivity

- JTRAIN1, Michigan manufacturing firms, 1988: $grant = 1$ for a job-training grant (19 of 54 firms). Grants were not random: q = unobserved productivity, possibly related to getting one.

$$\log(scrap) = \beta_0 + \beta_1 grant + \gamma q + v, \qquad q = \theta_0 + \theta_1 \log(scrap_{-1}) + r.$$

- Without the proxy: $\widehat{\log(scrap)} = \underset{(.241)}{.409} + \underset{(.406)}{.057}\, grant$, $R^2 = .0004$: the point estimate is positive but imprecise.

Back to reading the change in the schooling coefficient (4.3.2)

The lagged-outcome regression

Example 4.4: the proxy regression

with last year's scrap rate included, the grant effect changes sign and size

- With it: $\widehat{\log(scrap)} = \underset{(.089)}{.021} \underset{(.147)}{-.254}\ grant + \underset{(.044)}{.831} \log(scrap_{-1})$, $R^2 = .873$: a grant goes with a scrap rate about 22.4% lower than at a firm with the same rate last year ($t = -1.73$).

Exact effect $100[\exp(-.254) - 1] = -22.4\%$.

◂ Back to reading the change in the schooling coefficient (4.3.2)

◂ Back to the lagged-outcome setup

When the mean of q is not zero

Rewriting q around its mean shows the coefficient is still the APE

MODEL

$$y = \beta_0 + \beta_1 x_1 + \cdots + \beta_K x_K + \gamma_1 q + \gamma_2 x_K q + v, \quad E(v \mid \mathbf{x}, q) = 0$$

Let $\mu_q = E(q)$ and write $q = \mu_q + (q - \mu_q)$:

$$y = (\beta_0 + \gamma_1 \mu_q) + \beta_1 x_1 + \cdots + (\beta_K + \gamma_2 \mu_q) x_K + \gamma_1 (q - \mu_q) + \gamma_2 x_K (q - \mu_q) + v.$$

- This is (4.30) with the mean-zero unobservable $q - \mu_q$: the coefficient on x_K is $\beta_K + \gamma_2 \mu_q$.
- The APE is $E(\beta_K + \gamma_2 q) = \beta_K + \gamma_2 \mu_q$, the same number: assuming $E(q) = 0$ loses no generality.

Back to partial and average effects

Attendance effects can vary with prior GPA

The effect of attendance can vary with a student's prior achievement

$$stndfnl = \beta_0 + \beta_1 atndrte + \beta_2 priGPA + \beta_3 atndrte \cdot priGPA + \beta_4 priGPA^2 + u.$$

- **Partial effect**: $\beta_1 + \beta_3 priGPA$, evaluated at a student's prior GPA.
- **APE**: $\beta_1 + \beta_3 E(priGPA)$, averaged over the population.

Back to partial and average effects

Population and sample-mean centring

Only the intercept and the schooling coefficient move; the interaction and the fit do not

IQ entered as	IQ	$IQ - 100$	$IQ - \overline{IQ}$
educ	**.018**	**.052**	**.053**
	(.041)	(.007)	(.007)
IQ term	−.00094	−.00094	−.00094
	(.00516)	(.00516)	(.00516)
educ×IQ term	.00034	.00034	.00034
	(.00038)	(.00038)	(.00038)
R^2	.263	.263	.263

- $.052 - 100 \times .00034 \approx .018$: the same model, read at IQ 100 instead of IQ 0.
- Population mean unknown? Centre at the sample mean, here $\overline{IQ} = 101.3$.

‹ Back to centring the IQ proxy

A proxy with an interaction

Redundancy stays the same; sufficiency becomes a conditional-mean restriction

UNOBSERVED TERMS

$\gamma_1 q + \gamma_2 x_K q$

- **Redundancy**

$$E(y \mid \mathbf{x}, q, z) = E(y \mid \mathbf{x}, q). \tag{4.25}$$

- **Conditional-mean sufficiency**, with $E(q) = E(z) = 0$:

$$E(q \mid \mathbf{x}, z) = E(q \mid z) = \theta_1 z. \tag{4.34}$$

- Once z is known, $\mathbf{x}$ adds no information about the mean of q.

‹ Back to the meaning of the schooling coefficient › Replace the unobserved interaction

Replace the unobserved interaction

Include the proxy and its interaction with the observed regressor

PROXY ASSUMPTION

$E(q \mid \mathbf{x}, z) = \theta_1 z, \quad E(q) = E(z) = 0$

$$E(x_K q \mid \mathbf{x}, z) = x_K E(q \mid \mathbf{x}, z) = \theta_1 x_K z.$$

$$E(y \mid \mathbf{x}, z) = \beta_0 + \beta_1 x_1 + \cdots + \beta_K x_K + \gamma_1 \theta_1 z + \gamma_2 \theta_1 x_K z. \tag{4.35}$$

OLS includes z and $x_K z$; the coefficient of x_K estimates the APE, β_K.

‹ Back to the stronger proxy assumption · › Main: the schooling coefficient · › Centre the proxy at its mean · › Random coefficients and variance

Centre the proxy at its mean

Move the reference person from a zero proxy value to the population mean

UNCENTRED REGRESSION

$$E(y \mid \mathbf{x}, z) = a_0 + \sum_{j=1}^{K-1} \beta_j x_j + \delta x_K + az + \lambda x_K z$$

- The partial effect is $\delta + \lambda z$: the coefficient δ refers to $z = 0$.
- Let $\mu_z = E(z)$. Rewrite the same terms as

$$\delta x_K + \lambda x_K z = \underbrace{(\delta + \lambda \mu_z)}_{\text{APE}} x_K + \lambda x_K (z - \mu_z).$$

- Use $z - \mu_z$ and $x_K(z - \mu_z)$: the coefficient on x_K is now the APE.

‹ Back to replacing the unobserved interaction
› Main: the schooling coefficient
› Centring an observed interaction

Centre the observed interaction

Centring changes the coefficient's interpretation, not the fitted model

SAME MODEL

$$stndfnl = \alpha_0 + \alpha_1 atndrte + \beta_2 priGPA + \beta_3 atndrte(priGPA - \mu) + \beta_4 priGPA^2 + u, \qquad \mu = E(priGPA).$$

- Expanding gives $\alpha_1 = \beta_1 + \beta_3\mu$: the APE is one coefficient.
- Use a known μ, or the sample mean if it is unknown. ATTEND: sample mean of *priGPA* 2.59, and $\hat{\alpha}_1 = .0018$ (OLS se .0026).

‹ Back to centring the proxy › Main: the schooling coefficient

Random coefficients and heteroskedasticity

A random coefficient can induce heteroskedasticity when $\gamma_2 \neq 0$

CENTRED PROXY

$E(q \mid \mathbf{x}, z) = \theta_1 z$

- **Random coefficient** $y = \beta_0 + \cdots + \beta_K x_K + \gamma_1 q + \gamma_2 x_K q + v$ with $E(q \mid \mathbf{x}, z) = \theta_1 z$: assume $E(v \mid \mathbf{x}, q, z) = 0$, $\mathrm{Var}(v \mid \mathbf{x}, q, z) = \sigma^2$. Then

$$\mathrm{Var}(y \mid \mathbf{x}, z) = \sigma^2 + (\gamma_1 + \gamma_2 x_K)^2 \,\mathrm{Var}(q \mid \mathbf{x}, z).$$

  If $\gamma_2 \neq 0$, this can vary with x_K even if both component variances are constant.

Back to the average schooling return · Back to the average-effect assumptions

Appendix roadmap

APPENDIX F · Wooldridge 4.4 — current location in blue

Lagrange Multiplier Tests

Inference Illustrations

OVB, Selection and Proxies

Measurement Error

‹ Back to contents

Example 4.6: reported saving

The question is whether the reporting error moves with the regressors

$$E(sav^* \mid inc, size, educ, age) = \beta_0 + \beta_1 inc + \beta_2 size + \beta_3 educ + \beta_4 age.$$

- We observe reported saving sav, not actual saving sav^*.
- It may be reasonable that the error is uncorrelated with income, family size, education and age.
- Higher-income families may report more accurately.
- Different accuracy does not itself establish a biased slope.

‹ Back to reported scrap rates

Error uncorrelated with the recorded measure

This alternative assumption preserves consistency

COMPOSITE ERROR

$$y = \beta_0 + \beta_1 x_1 + \cdots + \beta_K x_K + (v - \beta_K e_K)$$

$$\text{Cov}(x_K, e_K) = 0. \tag{4.43}$$

- With the maintained assumptions, $v - \beta_K e_K$ is uncorrelated with every regressor: OLS is consistent.
- This differs from classical error, which assumes $\text{Cov}(x_K^*, e_K) = 0$.

Compare the two covariances | Neither polar assumption
Back to measurement error in a regressor

The covariance partner matters

Uncorrelated with the recorded value is different from uncorrelated with the truth

SHARED SETUP

$$x_K = x_K^* + e_K, \quad u = v - \beta_K e_K, \quad \text{Cov}(x_K, u) = -\beta_K \, \text{Cov}(x_K, e_K)$$

- **Recorded value**: $\text{Cov}(x_K, e_K) = 0$ implies $\text{Cov}(x_K, u) = 0$; OLS remains consistent.
- **True value**: $\text{Cov}(x_K^*, e_K) = 0$ implies $\text{Cov}(x_K, e_K) = \text{Var}(e_K)$; a nonzero slope is attenuated.

Back to alternative covariance Main: measurement error in a regressor

Nonclassical error has no universal direction

The direction depends on correlation with the true input

ONE REGRESSOR

$$x_1 = x_1^* + e_1, \quad u = v - \beta_1 e_1, \quad \text{Cov}(x_1, v) = 0$$

- Let $c = \text{Cov}(x_1^*, e_1) \neq 0$. Then $\text{Cov}(x_1, e_1) = c + \text{Var}(e_1)$.
- The scalar probability limit becomes

$$\text{plim}\, \hat{\beta}_1 = \beta_1 \left[1 - \frac{c + \text{Var}(e_1)}{\text{Var}(x_1)} \right].$$

Back to alternative covariance Main: measurement error in a regressor

Deriving scalar attenuation

Classical noise enlarges input variance without adding signal about the outcome

OBSERVED EQUATION

$$y = \beta_0 + \beta_1 x_1 + u, \quad u = v - \beta_1 e_1, \quad x_1 = x_1^* + e_1$$

$$\begin{aligned}
\operatorname{plim} \hat{\beta}_1 &= \beta_1 + \frac{\operatorname{Cov}(x_1, u)}{\operatorname{Var}(x_1)} \\
&= \beta_1 - \frac{\beta_1 \operatorname{Var}(e_1)}{\operatorname{Var}(x_1^*) + \operatorname{Var}(e_1)} \\
&= \beta_1 \frac{\operatorname{Var}(x_1^*)}{\operatorname{Var}(x_1^*) + \operatorname{Var}(e_1)}.
\end{aligned}$$

◂ Back to the attenuation factor

Responses to a mismeasured regressor

Improve the measure or obtain additional identifying information

- **Measure better**: reducing error variance brings the attenuation factor closer to one.
- **Interpret cautiously**: under classical error, the OLS probability limit is smaller in magnitude than the true slope.
- **Use additional information**: a valid second measure or instrument can identify the effect; this requires new assumptions developed in Chapter 5.

Back to the attenuation picture

Attenuation with several regressors

Controls absorb part of the signal but none of the noise

PROJECT THE TRUE INPUT ON THE OTHER REGRESSORS

$$x_K^* = \delta_0 + \delta_1 x_1 + \cdots + \delta_{K-1} x_{K-1} + r_K^*$$

$$\operatorname{plim} \hat{\beta}_K = \beta_K \frac{\sigma^2_{r_K^*}}{\sigma^2_{r_K^*} + \sigma^2_{e_K}}. \tag{4.47}$$

- r_K^*: the part not explained by the other regressors. More collinear: worse.
- NLS80, noise as variable as schooling: .060 → .031 alone; .065 → .030 with controls.
- The other coefficients are generally inconsistent too, in either direction.

▸ Deriving (4.47) ▸ Attenuation at every noise level ◂ Back to the attenuation factor

Deriving the multivariate attenuation formula

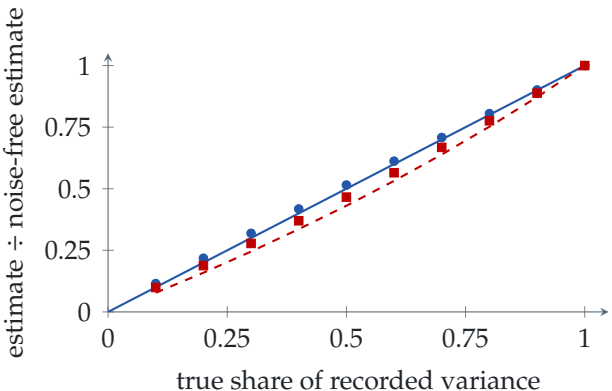
Partial out the other regressors, then repeat the one-regressor argument

- Partialling out: $\operatorname{plim} \hat{\beta}_K = \operatorname{Cov}(r_K, y) / \operatorname{Var}(r_K)$, where r_K is the error from projecting the recorded x_K on $1, x_1, \ldots, x_{K-1}$.
- Project the truth: $x_K^* = \delta_0 + \delta_1 x_1 + \cdots + \delta_{K-1} x_{K-1} + r_K^*$. Because e_K is uncorrelated with $1, x_1, \ldots, x_{K-1}$, the recorded input has $r_K = r_K^* + e_K$.
- $\operatorname{Cov}(r_K, y) = \beta_K \operatorname{Cov}(r_K^* + e_K,\, r_K^*) = \beta_K \sigma^2_{r_K^*}$, since $\operatorname{Cov}(e_K, r_K^*) = 0$ under classical error and v is uncorrelated with every regressor.
- $\operatorname{Var}(r_K) = \sigma^2_{r_K^*} + \sigma^2_{e_K}$. Dividing gives (4.47).

‹ Back to attenuation with controls › Return to the attenuation factor

Attenuation at every noise level

Controls push the estimates further below the one-regressor line



- NLS80: add noise to recorded schooling, re-estimate the log-wage slope.
- Alone: points on the line (4.48).
- With the six controls: below it.
- The dashed curve is (4.47).

‹ Back to attenuation with controls › Return to the attenuation factor

Practice: which way does the bias go?

Decide the direction, and whether classical error is believable, before discussing

1. Students report family income; regress college GPA on it and on high-school GPA and SAT.
2. The same, also controlling for parents' education and occupation.
3. Years of schooling reported in round numbers, in a wage equation.
4. Self-reported days of marijuana use in the last month, in a wage equation.
5. App users' self-reported age, in a spending regression.

◂ Back to the practical question